



Effect of cathode recirculation on high potential limitation and self-humidification of hydrogen fuel cell system

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HIGHLIGHTS

- A fuel cell system with cathode recirculation is built up based on a 30 kW stack.
- The single cell voltage is limited to less than 0.8 V at low currents.
- The cooperative working performance of different parameters is obtained.
- Relative humidity can be controlled with the recirculation ratio from 25 to 90%.
- The water removal function of the cathode circulation system was verified.

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ABSTRACT

The high potential at the start and stop of the stack, as well as the flooding and film drying phenomena that occur during operation, will seriously cause the performance degradation of PEMFC. In this study, the effects of the cathode cycle on the high-voltage limiting strategy and water management characteristics of the hydrogen fuel cell system under different operating conditions was measured using the orthogonal test and the AC impedance measurement methods. The experimental results show that by reducing the fresh airflow and increasing the amount of air recirculation, the oxygen concentration can be reduced and the single piece voltage can always be kept below 0.8 V under a small load. Air recirculation improves the relative humidity at the cathode inlet close to the humidifier, thereby improving the output performance of the stack and having better dynamic characteristics. Furthermore, air recirculation can also enhance the water removal capacity of the stack outlet under high power operation with less pump power, thereby significantly improves the consistency of the distribution of oxygen along the flow channel and increasing the durability of the stack. The conclusion can provide more specific guidance for fuel cell performance improvement and control optimization.

1. Introduction

With its advantages of green, extreme endurance and high energy density, hydrogen fuel cell vehicles are becoming an important application area of hydrogen energy and are becoming one of the main development directions of new energy vehicles [1]. At present, many companies and institutions have begun research on hydrogen fuel and development of fuel cell vehicles, with a focus on improving the power density of fuel cells and extending their service life [2]. As the core component of the fuel cell, the membrane electrode assembly (MEA) seriously affects the service life of the stack, which is one of the technical

challenges that restrict its commercialization [3]. When the proton exchange membrane fuel cells load changes, there will be a phenomenon of response lag, overshoot, etc., resulting in a high potential. The high potential (generally greater than 0.82 V) causes corrosion of the carbon support in the catalytic layer of the membrane electrode on the one hand, and catalysis of platinum on the other hand, reducing the service life of the fuel cell [4]. When the fuel cell operates under dynamic conditions such as frequent start-up, loading, load reduction, shutdown, etc., due to the sharp mutation of operating parameters (current density, gas flow, temperature, humidity, pressure, etc.), it is very easy to appear local drying, flooding, reaction gas loss and other phenomena [5].

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Therefore, it is very important to study the effects of high potential suppression methods and water management strategies on fuel cell performance.

Lin [6] proposed a two-phase auxiliary load strategy to avoid the hydrogen/air boundary, but severe local starvation was observed during the shutdown, and corrosion occurred in the entrance area. Moreover, high potential will cause the dissolution of platinum and its oxides, resulting in a decrease in the surface area of the catalyst [7]. Sharma [8] found through experiments that after 1500 times of 0.1–1.0 V potential cycle, the catalyst layer area of the fuel cell decreased by 40%, and the carbon corrosion rate of the catalyst layer increased with the increase of the upper limit potential, and the activity degradation of such catalysts during AST takes place. Dou [9] explored the anode catalyst carrier, and electrochemical tests showed that the electrochemical stability of Pt/SnO₂ under high potential electrochemical oxidation and potential cycling was significantly higher than that of Pt/C. Senoo [10] found that Pt-Ta-SnO₂-δ is more stable than commercial Pt/C in the high potential region and can be used as an ideal high-endurance cathode material for fuel cells. Under idling conditions, oxygen and hydrogen will pass through the proton exchange membrane to the anode and cathode, respectively. Hydrogen peroxide generated at the anode causes perforation of the membrane, causing hydrogen to flow directly from the anode to the cathode, thereby reducing the amount of proton exchange. While membrane decomposition leads to catalyst poisoning and reduced active area [11]. High potentials can also cause carbon corrosion of the bipolar plates, which can lead to severe corrosion of the coated metal plates, resulting in increased contact resistance and reduced stack output performance [12,13]. For the durability problem at high potential, it is simpler and more versatile to reduce the stack decay rate by the voltage clamping control strategy compared to the improved material performance [14]. In the load cycle test, Nissan compared the decay rate under voltage, showing that the voltage clamp can reduce the decay rate and the lower the voltage of the upper clamp, the more the attenuation rate reduced. Compared with no voltage clamping control strategy, the voltage decay rate can be reduced by 1/4 to 1/3 under the voltage clamp control strategy [15].

Proton exchange membrane can achieve efficient proton conduction only when fully wetted, so humidification is critical to the efficiency and performance of fuel cell systems. However, due to the parasitic power loss of the external humidifier, the system has high complexity, high cost and large volume; self-humidification is one of the key technologies for fuel cell research and development [16]. In recent years, some experiments and numerical simulations have conducted to study the effects of humidity on the transient performance of PEM fuel cells. Kim [17] tested the gas recirculation system in five fuel cell low-pressure systems and found that increasing the recirculation ratio increased intake air humidity and reduced O₂ concentration. Despite the use of a cathode recirculation structure, the dynamic characteristics of the system and the boundary conditions of the voltage clamping and the application of the control strategy to the engine are not studied. Zhang [18] studied a new cooling strategy for a PEM fuel cell stack using a three-dimensional multiphase non-isothermal model. Zhu [19] and Lin [20] found that the single variable cross section serpentine flow field has the largest pressure drop and flow rate, resulting in minimal ice formation during the cold start process. Kim [21] developed a two-phase flow model of a three-dimensional PEMFC complex flow field and added additional flow resistance around the baffle to improve interfacial liquid water and mass transfer. In addition, researchers also used the CFD method to quantitatively study the three-dimensional unsteady two-phase model on the cathode side to evaluate the effect of liquid water distribution on reactant transport and current density distribution [22]. The flow field design plays an important role in pressure distribution, current density, temperature, heat and water management, and Atyabi [23] found that the sinusoidal flow field has better drainage performance. S. Anyanwu [24] reveal that the novel hybrid sinusoidal channel designs are consistent in terms of quicker water removal under varying hydrophilic

wall conditions. Houreh [25] investigated the effect of obstacle shape and number on a membrane humidifier performance, and water transfer rate increases by 7.28% with one rectangular obstacle. Ramin [26] found that channels with two traps provide considerably higher current densities, and also improve the distribution of oxygen and water over the cathode catalyst, resulting in less cathodic overpotential across the cell. Kang [27] found that the water-removal capability of the ginkgo flow field was superior to that of the parallel and net leaf flow fields, but inferior to that of the serpentine flow field. Bao [28] found that the air guidance effect of 3-D baffles contributes to the reactant transport, and a separated liquid-gas transport phenomenon can be observed, which reduces the liquid coverage area on gas diffusion layer surface and provides a larger passage area for mass transfer. Yu [29] simulated the dynamic process of liquid water moving through the gas diffusion layer by a 3D lattice Boltzmann model, and two through-plane distribution styles were analyzed regarding their impact on the liquid water's flow behavior. It was found that there was a general dryness at the entrance of the fuel cell, resulting in uneven distribution of current density on the membrane electrode, shortening its service life. It is also feasible to achieve self-humidification by adding hydrophilic porous polyester fibers to the membrane and adding hydrophilic SiO₂ particles to the catalyst layer [30–34]. However, compared with numerical simulation, the experimental results are more reliable. Nowadays, there still lacks effective and feasible simulation methods for research work of fuel cell system quantitative analysis.

In the existing researches: most studies have quantitatively analyzed the trend of stack output voltage as inlet humidity changing, however, there is no better way to solve the problem of self-humidification than using a bulky humidifier; most studies elaborate on the attenuation caused by high potential during the start-stop process, but little attention has been paid to how to suppress the high potential in addition to reducing the start-stop times; Quantitative researches on the influence of cathode cycle on fuel cell high potential limit and self-humidification effect are still lacking.

Exhaust gas recirculation systems have been widely used and intensively studied on conventional fuel engines, but it is puzzling that there are no related applications on PEMFC systems [35–37]. Combined with the current research status, the life of the membrane electrode is the primary factor affecting the durability of the fuel cell, and the high potential affects the service life of the catalyst to some extent. In view of the above deficiencies, in order to reduce the single piece voltage of the fuel cell under low current density conditions and realize the self-humidification function at high power, a PEMFC system test bench using cathode circulation was designed by referring to the waste recycling system of the fuel machine [38]. The cathode circulation system is designed according to the generator theory and self-humidification strategy, and the effects of the amount of fresh air and the amount of circulation on the high potential limit of the monolithic voltage is quantitatively analyzed. Based on orthogonal experiments, the function of the circulating pump speed to suppress high potential was measured and qualitatively analyzed. The dynamic changes of humidification capacity and water content were measured with the cathode cycle, and the optimal combination under different load conditions and operating parameters was qualitatively analyzed. The humidification capacity of the cathode cycle was compared with that of an external humidifier, and the benefits of the cathode cycle for the removal of liquid water from the stack outlet were discussed in detail. The research results of this paper are based on the PEMFC cathode cycle system, which can provide theoretical basis for the design of high-performance, long-life PEMFC systems, and provide more specific guidance for control optimization.

2. Mechanism of cathode circulation

Due to the large amount of water produced by the fuel cell during operation, the cathode exhaust gas contains a large amount of high-temperature water vapor, which can be used to directly wet the intake

air. It can not only realize the self-humidification of the fuel cell, but also use the exhaust gas with a higher temperature to preheat the incoming air, to reduce the temperature difference distribution inside the stack. In addition, according to the Butler-Volmer equation, only the cathode over potential is considered, and the oxygen concentration is the ideal control variable among the controllable factors affecting the voltage. In order to make the fuel cell model based on cathode cycle is easy to use in the actual simulation and control system, make the following assumptions:

(1) When high temperature and humidity exhaust gas are mixed with low temperature and humidity air, it is assumed that there is no condensation in the mixing process, that is, the enthalpy value of fresh air and exhaust gas is equal to the enthalpy value of the mixed gas. Therefore, the concentration of the oxygen, nitrogen and water vapor components meet a simple linear relationship before and after mixing. (2) In the calculation process of the cathode cavity model, the penetration of nitrogen between the cathode and anode is not considered. (3) Assuming that the temperature of the exhaust gas after the air circulation pump remains unchanged, the relationship between the pressure ratio between the exhaust gas flow rate and the speed of the circulation pump in actual control can be obtained by querying the MAP chart.

According to the electrochemical theory, the cathode overpotential equation is as follows [39]:

$$\eta_c = V_{\text{cell}} + i_{\text{fc}}(R_{\text{ohm}} + R_{\text{con}}) - U_0^c \quad (1)$$

Where, V_{cell} is the average voltage, U_0^c is the cathode equilibrium potential, R_{ohm} and R_{con} are the ohmic and contact impedances, respectively.

Membrane water content [39]:

$$\sigma_m = \begin{cases} (0.5139\lambda - 0.326)e^{1268(1/303.15 - 1/T_{\text{fc}})}, & \lambda > 1 \\ 0.1879\lambda e^{1268(1/303.15 - 1/T_{\text{fc}})}, & \lambda \leq 1 \end{cases} \quad (2)$$

The cathode exchange current density is:

$$i_{\text{fc}} = \frac{s_0 - s_{\text{ref}}}{s_0} \left(\frac{C_{\text{O}_2}}{C_{\text{O}_2}^{\text{ref}}} \right) \exp \left(- \frac{\alpha_c}{RT_{\text{fc}}} F \eta_c \right) \quad (3)$$

Where, s is the liquid water saturation of the catalyst layer. $C_{\text{O}_2}^{\text{ref}}$ is the reference oxygen concentration.

In the cathode circulation system, the cathode inlet pipe is connected to the outlet end of the air circulation pump, and the inlet airflow of the inlet pipe is the sum of the airflow of the compressor and the air circulation pump:

$$W_{\text{to}} = W_{\text{fre}} + W_{\text{re}} \quad (4)$$

Where, W_{fre} is the fresh air mass flow rate, W_{re} is the recirculated air mass flow rate, and $W_{\text{sm_in}}$ is the total intake air mass flow rate after mixing.

The theoretical calculation formula of relative air humidity is as follows [40]:

$$\log p_{\text{sat}}(T) = 2.953 \times 10^{-2} \times T - 9.1837 \times 10^{-5} \times T^2 + 1.4454 \times 10^{-7} \times T^3 - 2.1794 \quad (5)$$

$$RH_{\text{theo}} = \frac{p_{\text{sat}}(T_{\text{ca_hm}})}{p_{\text{sat}}(T_{\text{fc}})} \quad (6)$$

It can be observed in Eq. (1) and Eq. (3) that the relationship between the voltage of fuel cell and oxygen concentration can be realized by reducing the oxygen concentration of intake air in order to reduce the voltage of the fuel cell. At the same time, the cathode circulation structure is designed to make full use of the large amount of water flowing from the exhaust port of the cathode. A gas-circulating pump is used between the cathode outlet and the inlet of the stack to pump gas

and water vapor from the stack outlet to the inlet, and a stop valve is installed at the cathode outlet to control the amount of circulating gas. In addition, a hand valve is also installed behind the air compressor to limit the amount of fresh air during circulation, and the specific structural principle is shown in Fig. 1(a). Through the recycling of cathode gas, the concentration of oxygen in the intake air is reduced, and self-humidification is realized at the same time, which has useful application value. In order to obtain the change rule of the characteristics of the cathode cycle fuel cell system, a test system is built, which includes hydrogen system, air system, cooling system and other subsystems, as shown in Fig. 1(b), and the configuration parameters of the stack used in the test system are shown in Table 1.

3. Discussion

3.1. Voltage limiting effect with cathode circulation

According to the gradient step loading of 40A, the maximum and minimum are 400A and 40A, respectively; The time interval of each load is about 100s, and the total test time is 2000s. It can be viewed in the dynamic response test result of the fuel cell voltage shown in Fig. 2 that the voltage gradually decreases when the current is applied, and the voltage gradually increases when the load is reduced. In addition, the response time and minimal regulation of the fuel cell are not the same under the same step current. In order to reduce the degradation of the membrane electrode performance caused by high potential, the voltage should be limited to less than 0.8 V during the start-stop phase. Therefore, when the stack current is less than 100A, the monolithic voltage is more than 0.8 V, which is a range of limited voltage.

After the circulation pump is turned on, the exhaust valve of the outlet is closed and the opening of the intake valve is reduced, so that the exhaust gas and a small amount of fresh air is mixed before the cathode inlet and then enter the stack together. After a gradient test time of 800 s, the oxygen concentration at the cathode inlet can be gradually reduced from the initial 18.6%–11%, and continues to decrease with the increase of the circulating pump speed until it drops to 3%, as shown in Fig. 3a. This is because as the circulating pump speed increases, the exhaust gas recirculation rate continues to increase, resulting in a continuous decrease in the oxygen concentration at the cathode inlet. Subsequently, due to the low oxygen concentration of the stack, the control system is forced to shut down, so the oxygen concentration must have a certain range limit, preferably not less than 5%. When the stack operates at low current as shown in Fig. 3b, if the air intake valve is always in full open state, the change of average voltage per slice will be very small even if the amount of circulating air is increased, and the cathode circulation system will not play a role of limiting potential. As the circulation pump is turned on, the exhaust gas is mixed with fresh air before the cathode inlet, the oxygen concentration at the cathode inlet decreases from the beginning, and the smaller the opening of the intake valve, the lower the oxygen concentration, as shown in Fig. 3(c). This is explained by the fact that the smaller the opening of the intake valve is, the less the amount of fresh air is, so the higher the proportion of the amount of recycling is. After the internal circulation, the concentration will reach a balance state, and the single cell voltage will also reduce, but the change of voltage has obvious delay. The change in oxygen concentration at the cathode outlet throughout the process was so great that when the valve opening was 0.8, 0.6 and 0.4, respectively, the corresponding oxygen concentration decreased by 16%, 14% and 11%. It can also be seen that the intake valve opening degree is changed from 0.8 to 0.4, the single cell voltage is lowered from 0.9 V to 0.8 V, and as the fresh air excess coefficient is lowered, the rate of voltage drop becomes large. A rapid drop in voltage indicates that the change of oxygen concentration at the cathode inlet is less affected by the air recirculation system, while it is more affected by the fresh air excess coefficient, which is mainly due to the influence of the fresh air excess coefficient on the proportion of recirculation flow to the total cathode intake. Fig. 3d

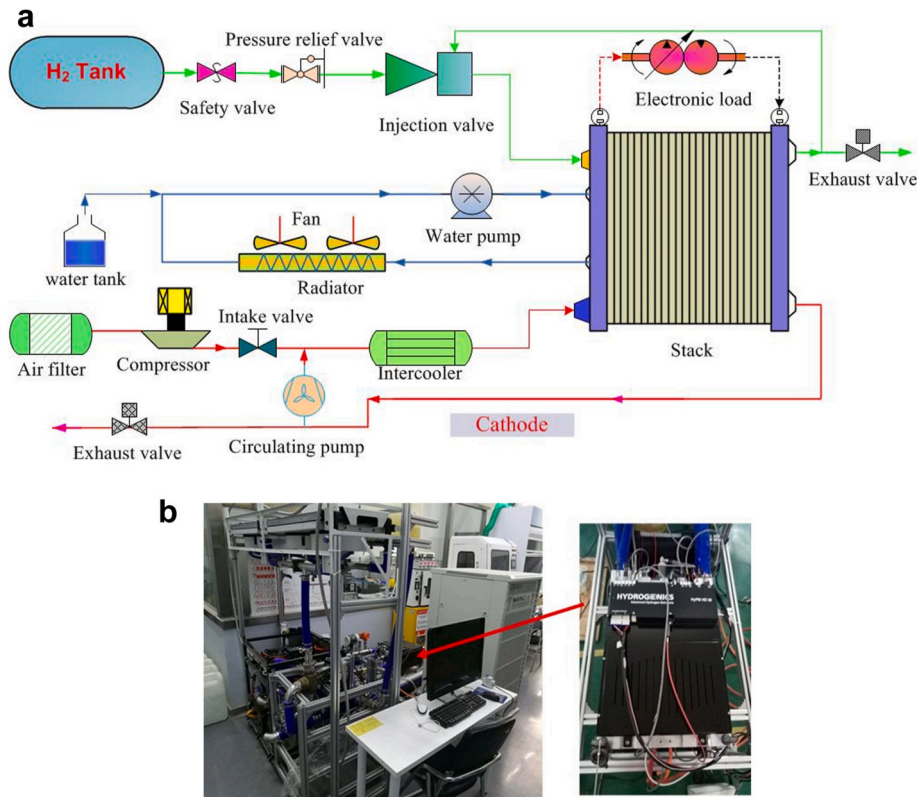


Fig. 1. Fuel cell test system with cathode cycle.

Table 1
Stack parameters and bench configuration parameters.

Parameter name	Parameter value
Stack power/kW	30
Active area/cm ²	500
Number of pieces	100
Stack operating temperature/°C	60
Voltage range/V	70–120
Current range/A	0–500
Gas channel width/mm	2
Gas channel depth/mm	0.6
Ridge width/mm	0.6
Membrane thickness/ μ m	30
Oxygen concentration sensor	0–21%, $\pm 0.5\%$
Temperature and humidity integrated sensor	–20–100 °C, $\pm 1.5\%$
Flow Sensors	0–100%, $\pm 1.5\%$

shows the voltage of 50 pieces of the stack measured by the voltage inspection module, in order to verify the practical application effect of the voltage limiting strategy. In addition, it can be seen that the single cell voltage is controlled below 0.8 V during the startup and shutdown stages, which achieves the purpose of cathode cycle, but the consistency between the single chips is slightly different. The single cell voltage is relatively stable when operating in a stable state, so the cathode cycle does not cause adverse effects in the stable operation stage of the stack.

When implementing the control method of the voltage clamp, reducing the fresh airflow rate and adjusting the cathode recirculation rate at the same time are important strategies to achieve a single piece voltage drop and maintain stable operation of the fuel cell system. However, although reducing the amount of fresh air alone can achieve the purpose of reducing the voltage drop, it will lead to a weakening of the diffusion capacity of oxygen to the reaction area under a small current density, which may easily lead to uneven gas distribution and local deficiency gas. Moreover, because the cathode intake airflow is reduced, the ability to take liquid water out of the stack is further

weakened, which easily leads to problems such as local flooding. In order to study the interaction of the characteristic parameters such as current, voltage, and reaction gas under the voltage clamping strategy, corresponding voltage clamping experiments were carried out, and the experimental results of 9 groups of orthogonal experiments are shown in Table 2. As can be seen, the effect of the fresh air excess coefficient on the oxygen concentration is obvious because the exhaust gas and fresh air are mixed before the cathode inlet; the excess coefficient of fresh air affects the ratio of recirculation flow to the total cathode intake. As the speed of the circulating pump increases, the exhaust gas recirculation rate continues to increase, resulting in a continuous decrease in the oxygen concentration at the cathode inlet. The oxygen concentrations of different fresh air excess coefficients were obtained through experiments, so that the voltage drop could be precisely adjusted and controlled under different current densities.

3.2. Self-humidifying effect with cathode circulation

In order to obtain the humidification effect of the system, it is necessary to explore the qualitative relationship between humidity and high frequency impedance. The dynamic experiment of the cathode circulation system was carried out by using the high-frequency impedance tester to monitor the high-frequency impedance of the membrane in real-time. It should be noted that according to the fuel cell system test experience, the dynamic time of humidity is usually longer, about 20s. According to a large number of theoretical and experimental results, high-frequency impedance is an effective indicator of membrane water content, and voltage has a good correspondence with high-frequency impedance, that is, high voltage at low impedance, low voltage at high impedance, and basically no delay, indicating that high frequency impedance is a good real-time observation tool for fuel cell systems. Fig. 4a shows the corresponding relationship between cathode inlet relative humidity, average single cell voltage and high-frequency impedance during the experiment, which can be seen that the cathode

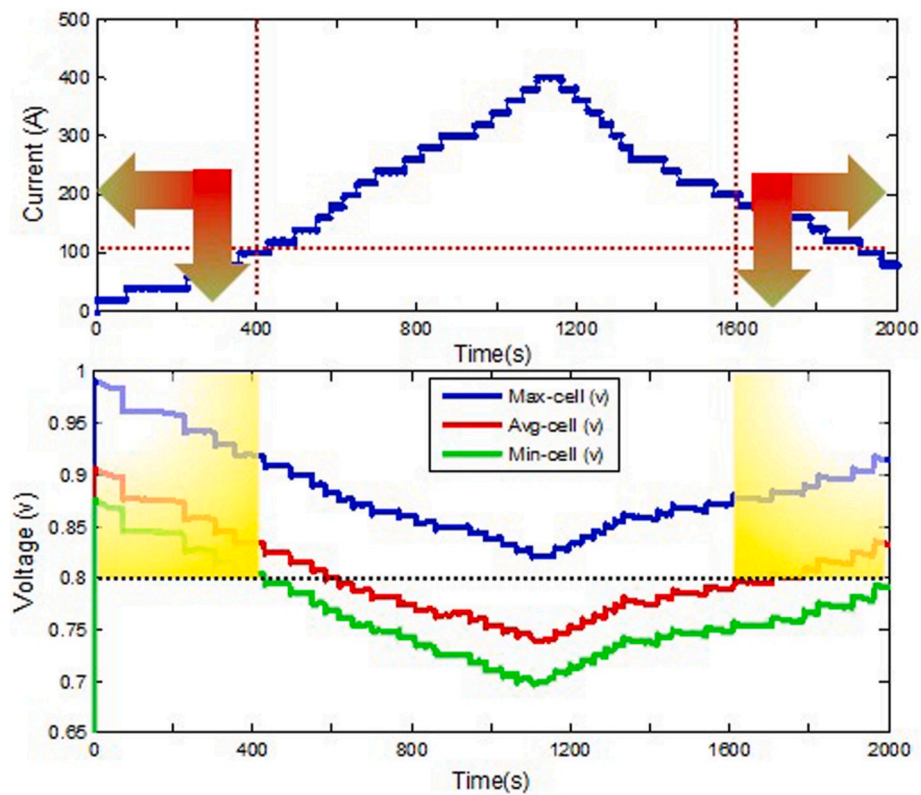


Fig. 2. Loading current and average voltage.

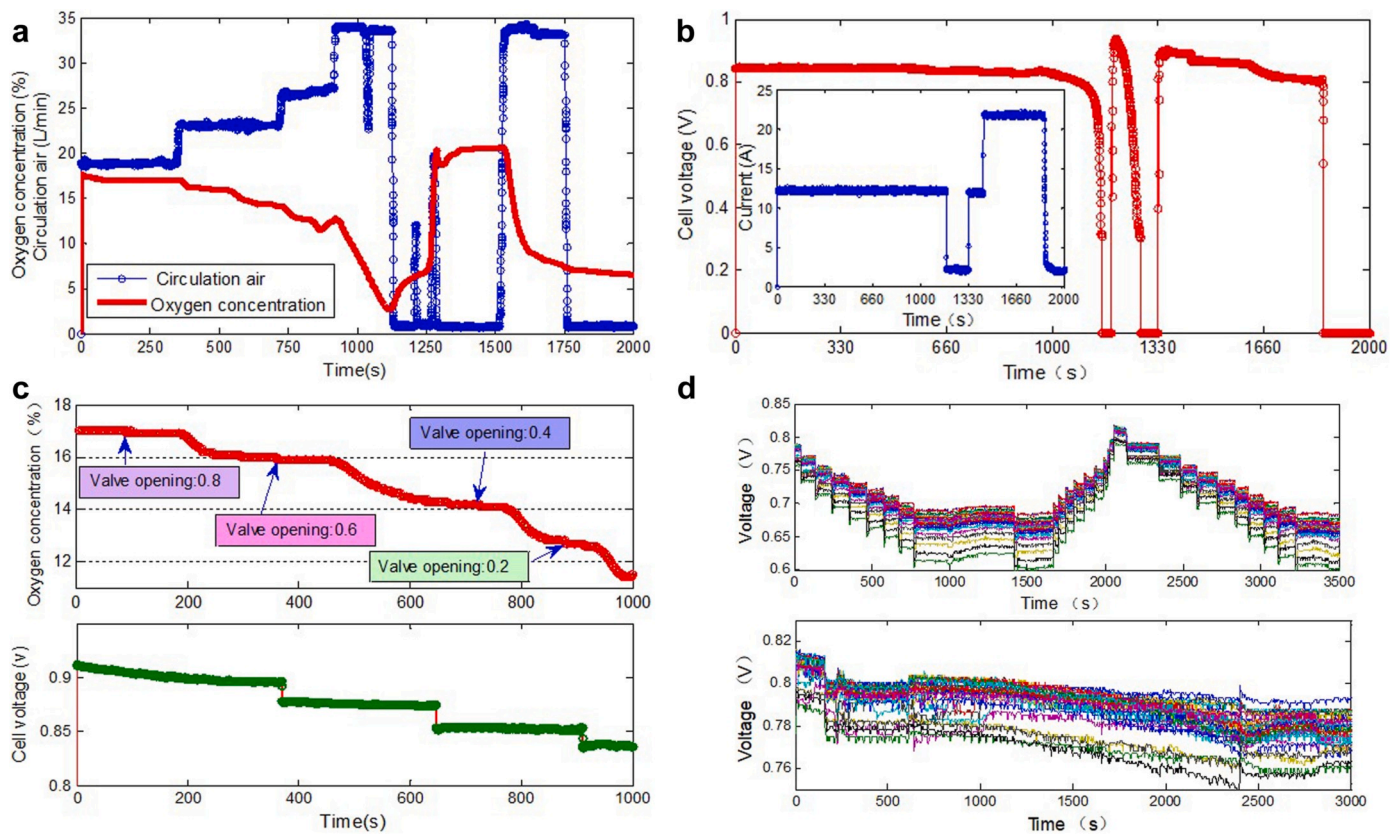


Fig. 3. Experimental results of voltage clamping.

Table 2
Experimental results of voltage clamping.

Case	Air excess factor	Current density (A/cm ²)	Circulating pump speed (rpm)	Inlet oxygen concentration (%)	Outlet oxygen concentration (%)	Cathode inlet humidity (%)
1	3	0.12	1500	15.6	13.8	72
2	2.5	0.15	2000	13.6	10.5	83
3	1.5	0.18	2500	9.8	5.6	85
4	3	0.2	2500	15.6	13.1	80
5	2.5	0.22	1500	14.6	10.2	85
6	1.5	0.28	2000	10.3	8.4	90
7	3	0.15	2000	16.3	13.5	85
8	2.5	0.25	2500	13.3	11.9	68
9	1.5	0.28	1500	10.1	5.8	93

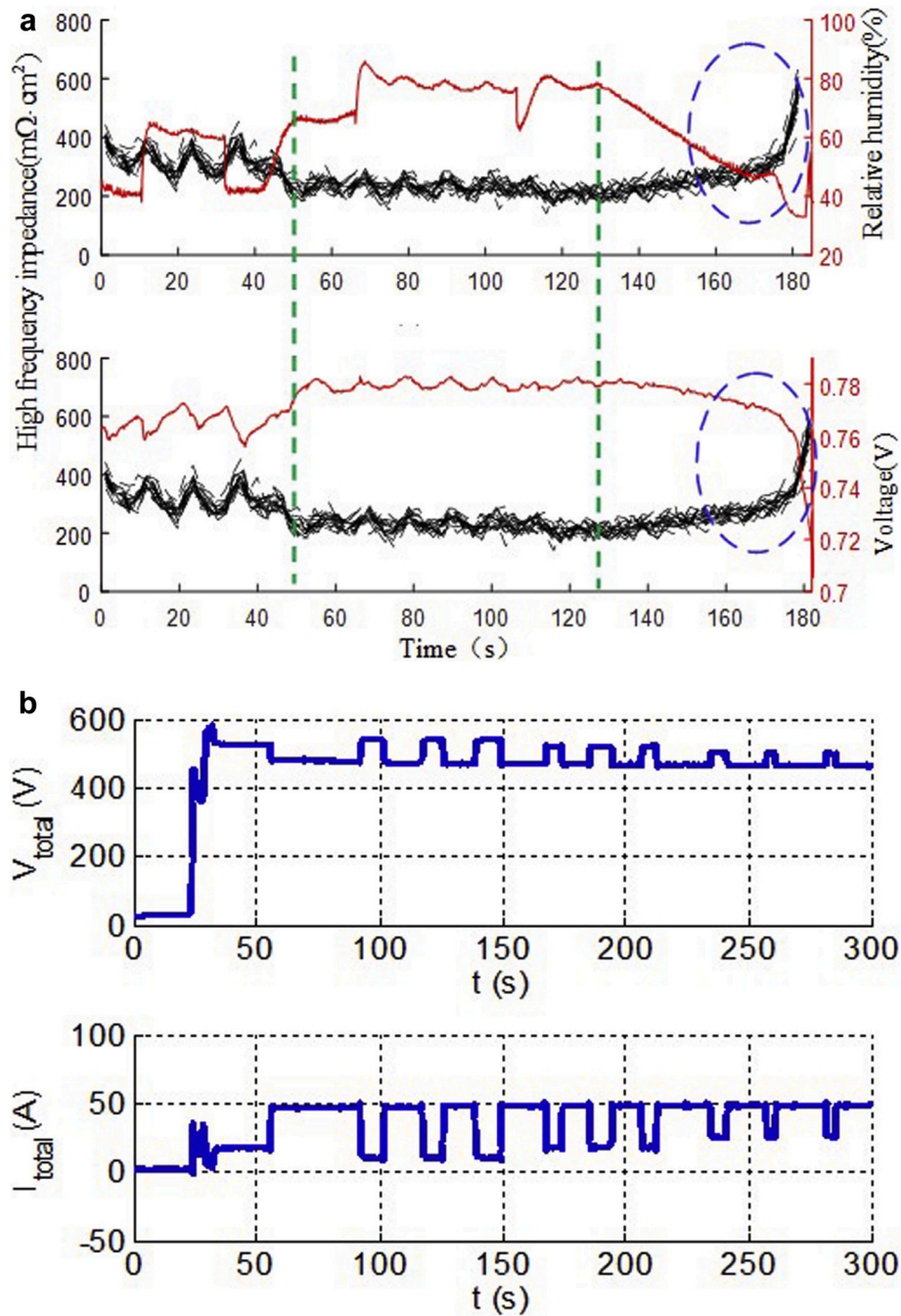


Fig. 4. Experimental results.

inlet humidity response time caused by the air circulation pump speed step is very rapid, basically within 20s. During the rapid increase of the membrane impedance from 300 to 500 mΩ cm², the average monolithic voltage was reduced from 0.78 V to 0.70 V, and the performance attenuation was about 10%. Moreover, since the single cell voltage drops almost linearly after 180s of the test time, it can be established that the voltage has been unstable at this time. Thus, it can be concluded that when the high-frequency impedance is in the range of 200–400 mΩ cm², it corresponds to the ideal working state of the stack. The current step sequence of 51A->12A, 51A->19A, 51A->27A was used, and each cycle was performed as shown in Fig. 4b. The test results show that the voltage follow-up effect is better, and the voltage fluctuation of the three cycles is smaller, indicating that the dynamic response characteristics of the stack are better. In addition, the average internal resistance of the fuel cell tested by the current step method is shown in Table 3, which can be used to predict the degree of humidification of the new system.

The comparison between theoretical and experimentally measured relative humidity at the cathode inlet is shown in Fig. 5(a). It can be observed that they have a good agreement with maximum error only 10%, indicating that the experimental results have better reliability. After a large number of experiments, it was found that the measured value of the humidity sensor would be higher than the actual value, mainly because there would be residual water droplets on the sensor, which would affect the measurement accuracy. If the effects of residual water were excluded, the error would be even smaller. Additionally, the experimental results show that the relative humidity of the cathode inlet for cathode circulation is over 70%, indicating that the self-humidification capacity of the cathode circulation system is close to the humidification performance of an external humidifier, which can significantly increase the relative humidity of the air at the cathode inlet. When the opening of the inlet valve is fixed and the exhaust valve is closed, the relative humidity at the cathode inlet is significantly affected by the exhaust gas re-circulation (EGR) rate. As the speed of the circulating pump increases from 1500 rpm to 3000 rpm, the relative humidity of the mixed air at the cathode inlet also rises slowly from the initial 20%–80%, as shown in Fig. 5(b). From the circled position, it can be seen that every time the rotation speed of the circulating pump is changed, the humidity will have an obvious beating and upsurge process, for the main reason that the strong air flow will carry a large number of liquid water at the outlet to the inlet at the moment of changing the rotation speed, resulting in the instantaneous increase of humidity. When liquid water slowly becomes less, the humidity slowly decreases to a certain extent and then slowly rises. It shows that the air recirculation system can significantly affect the humidification capacity of the system and achieve the purpose of system design. It is worth noting that although the circulating pump will generate an additional power consumption of up to 400 W, the system is simplified and the cost is greatly reduced. Compared with the widely used anode recycling [41], the humidification effect of cathode recycling is much better, which can improve the performance, especially in the case of low relative humidity. Shao [42] also proved this conclusion through simulation study.

In order to further study the characteristics of the fuel cell system under the self-humidification strategy, an orthogonal experiment of self-humidification as shown in Table 4 was carried out, and the results of

nine examples were obtained. Under high current density, the relative humidity rises rapidly from a minimum of 40%, and eventually reaches about 90% as the speed of the circulating pump reaches 2500 rpm, indicating that the air recirculation system can significantly affect the humidification capacity of the system. Furthermore, the high-frequency impedance decreases continuously with the increase of the rotation speed of the circulation pump, which indicates that air recirculation can increase the water content of the membrane and reduce the membrane impedance. This is because under high current density, the tail gas contains a large amount of water vapor, so the water concentration carried by the recirculated air is higher, and at the same time, the cathode inlet flow rate increases due to recirculation, especially at a large recirculation rate. The cathode inlet pressure will increase accordingly, and the partial pressure of water vapor will increase, further increasing the relative humidity at the cathode inlet, even reaching a relative humidity of 90%. At small and medium current densities, the increase in inlet relative humidity and the increase in pressure caused by a large airflow will reduce both ohmic loss and mass transfer loss, and improve stack performance.

Based on the data of orthogonal experiments, the parameter matching results of the cathode cycle system that can be used in practical control systems are obtained, as shown in Fig. 6(a). From the results of the dynamic change of airflow under different powers in Fig. 6(b), it is found that after the cathode cycle is adopted, the phenomenon of up-regulation will occur at each loading, but it will return to a stable state after a short time. In addition, according to the gas flow rate, the excess coefficient of air at different powers in Fig. 6(c) can be obtained, which indicates that the larger the flow rate of air, the greater the power of the stack, and the excess coefficient decreases accordingly. In addition, the cathode circulation increases the pressure of the gas entering the stack to a certain extent, and promotes the transmission of the gas to the diffusion layer. In addition, increasing the cathode inlet pressure can significantly improve the output performance, especially when the relative humidity of the cathode is low [43]. At the same time, the flow rate of the gas will increase, which will help quickly remove the generated moisture, thereby increasing the power of the stack. It is worth noting that the excess factor can also be reduced by adjusting the speed of the air compressor to prevent excessive gas flow.

3.3. Effect of cathode circulation on forced water removal

Another advantage of air recirculation is that it can enhance the migration of water at the outlet under high power operation, thereby significantly improving the consistency of the distribution of oxygen along the flow channel and increasing the durability of the system. According to existing research literatures [44,45], when the fuel cell is operated at a high current density, a large amount of liquid water will accumulate near the exit of the stack, causing flooding, which will affect the output of the stack. To illustrate the fuel cell voltage inconsistency, the standard deviation method shown in formula(7) is used in the statistics. Where, μ is the average value and n is the number of data points [46].

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} \quad (7)$$

Since it is usually necessary to combine hundreds of fuel cells into one stack in practical use, the service life of the stack is not only related to the service life of a single fuel cell, but also the difference that seriously affects the service life of the stack due to the inconsistent self-discharge rate. According to Eq. (7), the standard deviation of the monolithic voltages at 50A and 200A is 15 and 19 mV, respectively, indicating that the voltage consistency of the stack is good. However, when the current is increased to 400 A, the standard deviation of the voltage is calculated to be 25 mV exceeding the allowable limit of 20 mV, which means that the consistency is poor, as shown in Fig. 7(a).

Table 3

Current step test data.

U ₁	U ₂	ΔU	I ₁	I ₂	ΔI	R/Ω
484	528.6	44.6	44.81	16.33	−28.48	−1.566
489.6	516.5	26.9	38	22.21	−15.79	−1.704
476.1	512.3	36.2	45.96	24.07	−21.89	−1.654
473.3	517.4	44.1	46.24	19.13	−27.11	−1.627
475.9	509.8	33.9	43.32	23.12	−20.2	−1.678
469.2	512.2	43	46.93	21.49	−25.44	−1.69
467.8	498.5	30.7	47.61	28.87	−18.74	−1.638
468.3	499.1	30.8	47.26	27.79	−19.47	−1.582

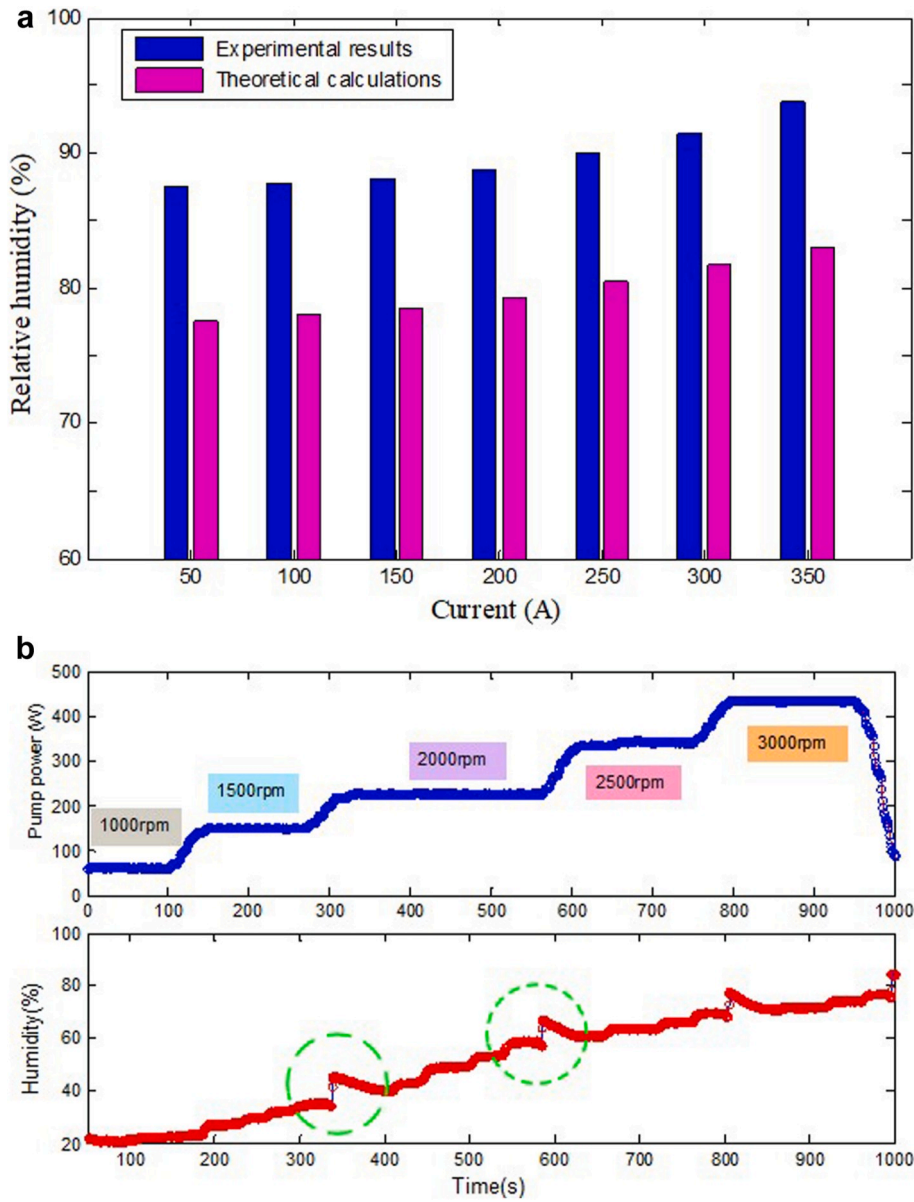


Fig. 5. Self-humidifying capacity of cathode circulation system.

Table 4
Self-humidification experiment results.

Case	Cathode humidification temperature	Circulating pump speed (rpm)	Current density (A/cm ²)	High frequency impedance (mΩ·cm ²)	Cathode inlet humidity (%)
1	30	1500	0.5	340	59
2	30	2000	0.7	290	65
3	30	3000	0.7	330	72
4	40	2000	0.7	300	59
5	40	2500	0.7	280	75
6	40	3000	0.9	250	80
7	50	2500	0.6	280	84
8	50	2500	0.8	230	88
9	50	3000	0.8	210	90

Under the current of 400A, many of the generated water vapor will condense into liquid water in the cathode flow channel due to the large output power of the stack, which hinders the gas transmission, resulting in lower voltage and large fluctuation of the 27th and 33rd pieces. Fig. 7

(b) also shows the steady-state voltage change of the 33rd fuel cell, indicating that the fluctuation of the single cell voltage becomes more and more noticeable as the output current of the stack increases. The main reason may be that the location is flooded due to too much liquid water, or the fuel cell is less able to withstand the pressure fluctuations inside the stack under high current. The test results show that the voltage inspection device can detect the all the single cell voltage changes in real time, and timely find the fuel cells with mediocre performance. It can be seen from Fig. 7(c) that when the current continues to run stably at 450A, the water content of the catalyst layer will reach the maximum due to the appearance of liquid water in the cathode catalyst layer, and the water content of the membrane at this time is related to the current density. The comparison of the experimental results shows that the uniformity of the single cell voltage is improved after the cathode cycle is used, and the voltage is also improved, thereby increasing the output power of the stack.

The air recirculation system can increase the relative humidity at the cathode inlet and the moisture content of the membrane, thereby improving the output performance of the stack. It can be seen from Fig. 8 (a) that the PEMFC system using the cathode cycle can not only achieve

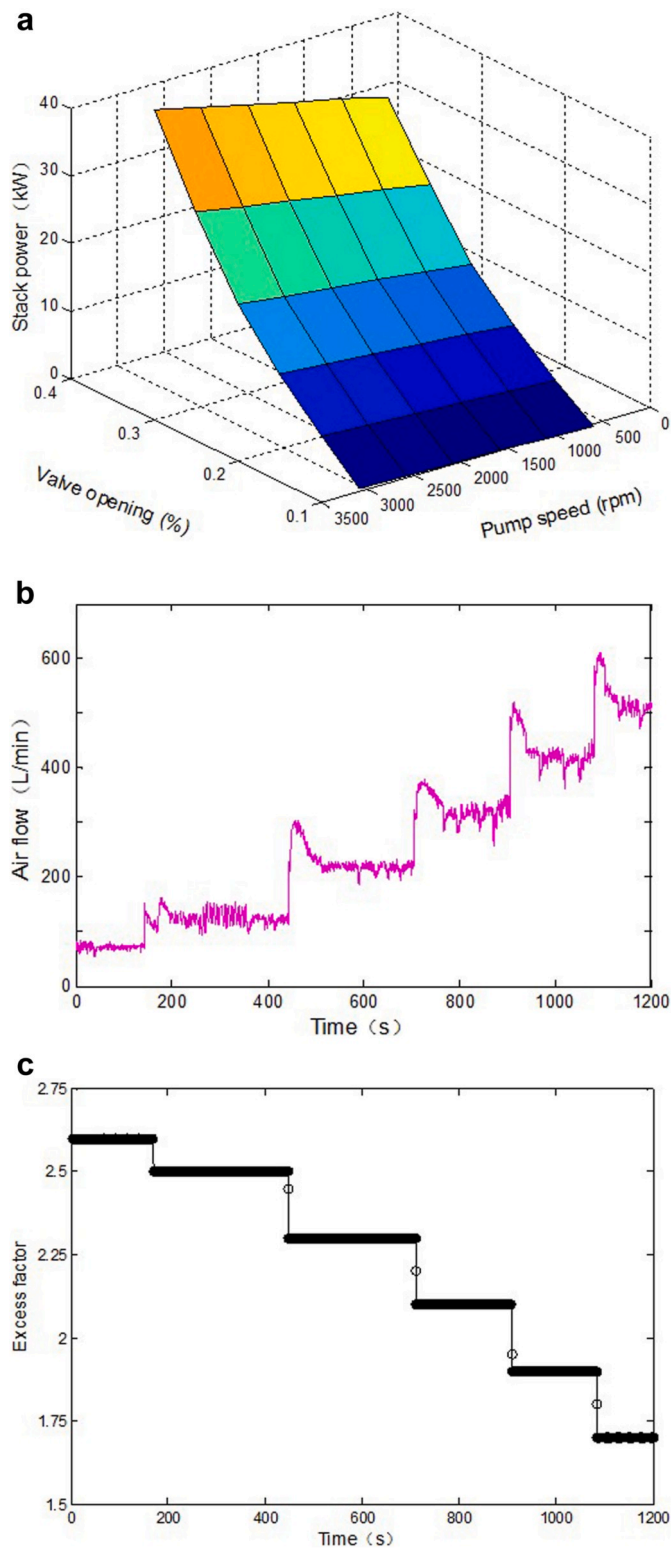


Fig. 6. Effect of cathode cycle on intake system.

the effect of using an external humidifier, which shows that the self-humidification performance has a considerable humidification capacity and effect compared with an external humidifier. At the same time, after 800s, the average single cell voltage decreases with a small current density and the voltage without humidification is significantly lower. As shown in Fig. 8(b) and (c), because the use of an external humidifier is likely to cause severe flooding downstream of the flow channel, the

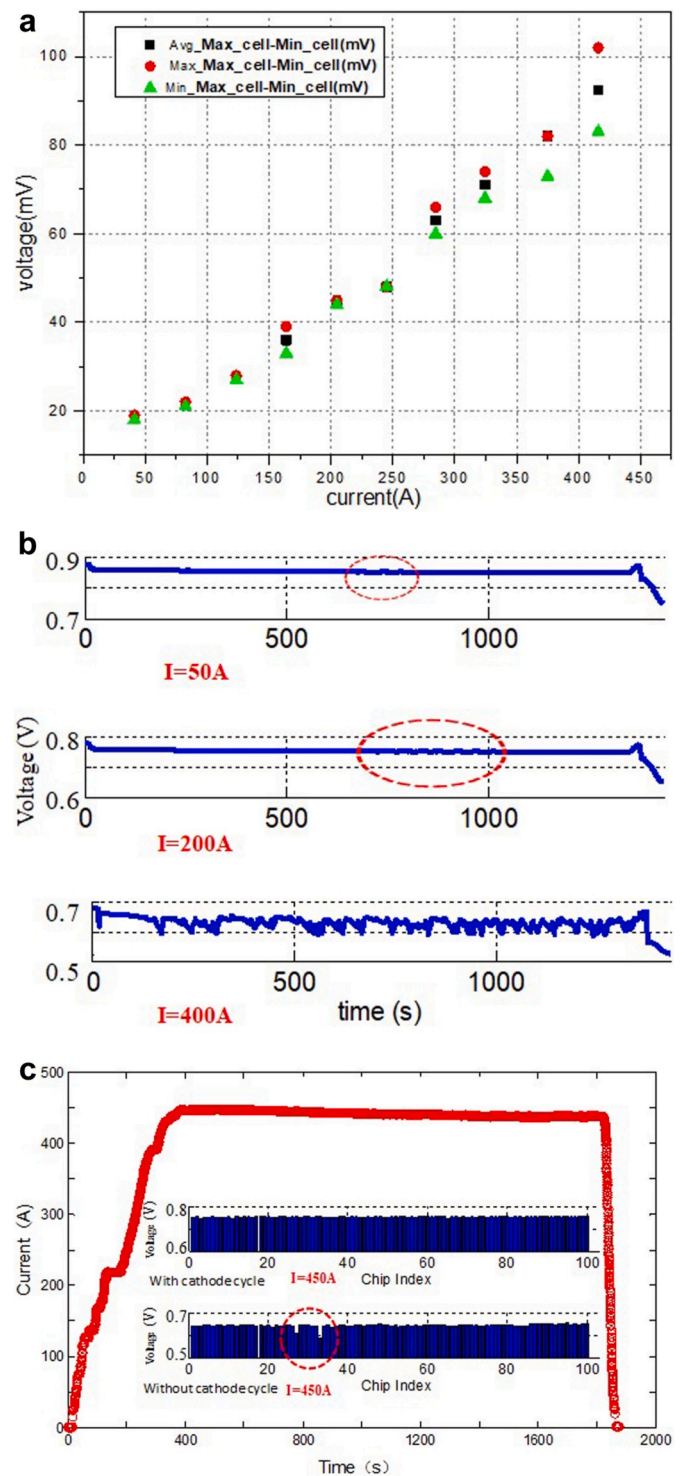


Fig. 7. Effect of cathode circulation on forced water removal.

power at high current density is lower than that using a cathode cycle, which confirms the enhanced water removal capacity of the cathode cycle. The main reason is that as the current density increases, the saturation of the liquid water inside the diffusion layer also increases, and the resistance of oxygen to the catalyst layer becomes larger. Therefore, the difference in power at high current density is caused by an increase in water concentration.

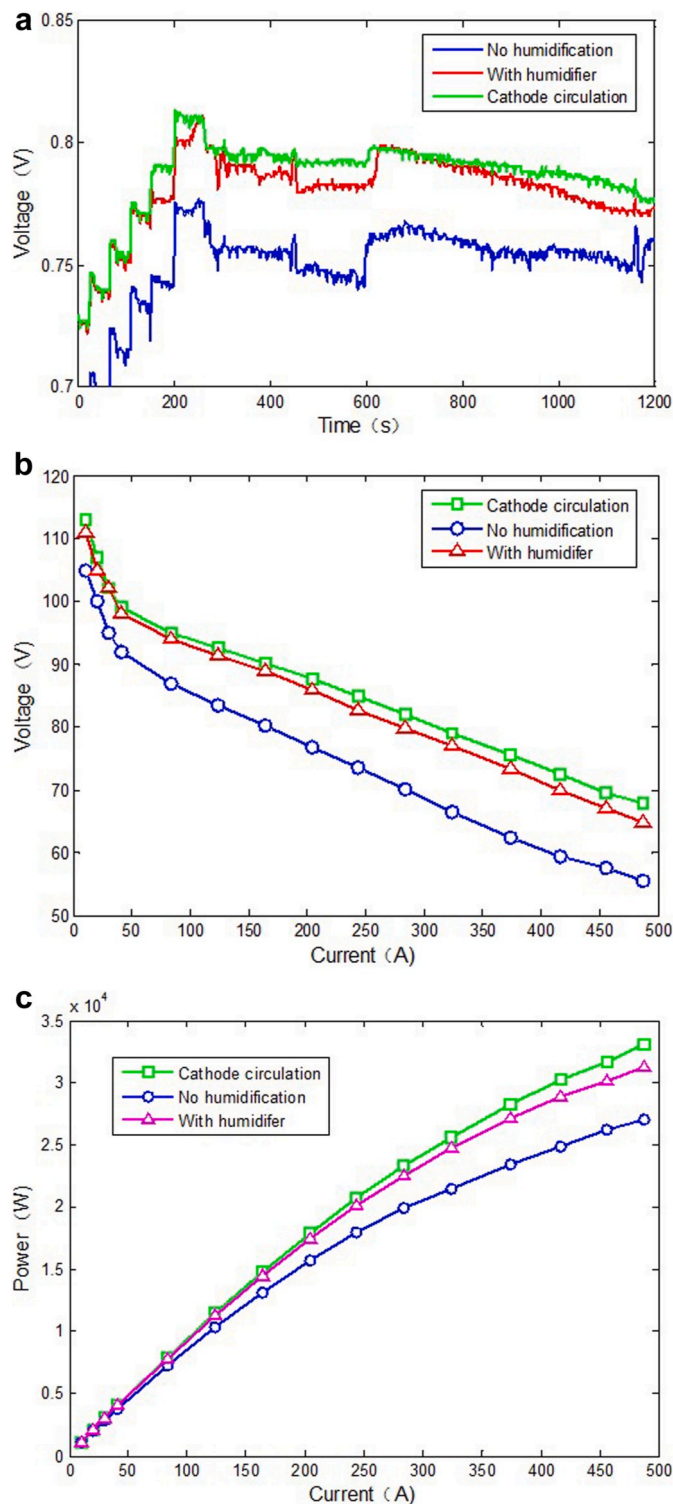


Fig. 8. Effect of cathode cycle on stack.

4. Conclusion

In this work, a hydrogen fuel cell system test bench with recycling

Nomenclature

i	Current density [A/cm^2]
I	Current [A]

cathode was built with which the orthogonal experiments of voltage clamping were carried out in order to improve the working performance and prolong the service life of the stack. A series of experiments are conducted on a stack to investigate the effects of this mode on the operating characteristics of fuel cells. The experimental results show that:

- (1) An innovative proposal is to propose a cathode cycle to limit the high potential that occurs during startup and shutdown to reduce the damage to the life of the stack. Constant current loading dynamic measurement found that when the stack current is below 100A, high potential limiting measures are required. By adjusting the opening of the exhaust valve, a coordinated control of the fresh air and recirculated gas flow is achieved, thereby reducing the oxygen concentration of the mixed gas at the inlet of the stack, and the single piece voltage can always be controlled below 0.8 V at low current density regions.
- (2) The AC impedance test shows that the humidity of the inlet air has a direct effect on the water content and the output voltage of the membrane, and the impedance in the range of 200–400 $m\Omega cm^2$ corresponds to the optimum humidity. The relative humidity of the cathode inlet for cathode circulation is over 70%, indicating that the self-humidification capacity of the cathode circulation system is close to the humidification performance of an external humidifier.
- (3) The cathode recirculation mode can not only avoid membrane dryness caused by too little water generation at low current density without external humidifier, but also improve the water removal capacity of the stack outlet at high power. Consequently, the water balance between the single pieces is improved at high current density regions, so the consistencies of the single cell voltage are improved.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Qinguo Zhang: Data curation, Writing - original draft. **Zheming Tong:** Supervision, Writing - review & editing. **Shuiguang Tong:** Conceptualization, Methodology.

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C	Substance concentration [mol/m ³]
T	Temperature [°C]
P	absolute pressure [Pa]
R_{ohm}	ohmic resistance [Ω]
F	Faraday constant [96487 C/mol]
G	Gibbs Free Energy [J/kg]
V_{cell}	monolithic voltage[V]
R_{con}	contact impedances[Ω]
M	mass flow rate [kg/m ³]
S_o	Liquid water saturation

Greek letters

λ	water content
η	over potential [V]
μ	average value[mV]

Superscripts and subscripts

<i>con</i>	contact
<i>C</i>	Cathode
<i>RH</i>	Relative humidity
<i>re</i>	recirculated air
<i>theo</i>	theoretical
<i>fre</i>	fresh air
<i>sat</i>	saturation
<i>out</i>	outlet
<i>in</i>	Inlet
<i>ref</i>	reference
<i>fc</i>	fuel cell
<i>to</i>	total

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